

Rooftop Solar Assessment Report

Source report: SolarTwin_Report2.pdf

Prepared for: Homeowner/Business Owner, Airoli, Navi Mumbai, Maharashtra

1. Executive Summary

This report evaluates the solar potential of a rooftop located at Airoli Station Road, Airoli Sector 2, Navi Mumbai, Maharashtra. Based on the automated assessment, the site can support a small photovoltaic (solar panel) system with a capacity of **2.75 kW** — enough to generate an estimated **4,097 kWh of electricity per year** (about 11.2 kWh per day). This is projected to save approximately **■32,775 per year** on electricity bills, with the system paying for itself in about **5 years**. Over a 25-year period, the estimated cumulative profit is **■654,378**.

The rooftop is rated as suitable for solar installation, though two physical obstructions on the roof were detected that slightly limit how much of the space can be used. The report recommends further on-site engineering checks before construction begins.

2. Site & Roof Assessment

- **Location:** Airoli, Navi Mumbai, Maharashtra — a coastal, tropical region that generally receives strong year-round sunlight, though with a pronounced monsoon season.
- **Usable Roof Area:** 17.02 m². This is the portion of the roof identified as clear and structurally suitable for mounting panels — it excludes water tanks, staircases, vents, or other rooftop clutter. For context, a usable area of this size is fairly modest, more typical of a small residential rooftop section than a full commercial roof, which explains why the recommended system here is small (2.75 kW) despite the report also describing it in places as "industrial-scale" — that phrase likely reflects the panel layout style rather than the actual system size.
- **Rooftop Obstacles:** The assessment identified **2 obstacles** on the roof (e.g., could be things like water tanks, exhaust vents, or shadows from adjacent structures — the source data doesn't specify which). These reduce the space available for panels. The report notes that **relocating these obstacles could allow for a larger system**, which is worth discussing with a roofing contractor if feasible.
- **Panel Orientation:** The system is laid out in **"Portrait"** orientation, meaning each panel is mounted in a vertical (taller-than-wide) position. This orientation was chosen because it packs panels more efficiently into the available space — in general, orientation choice is driven by roof shape and obstacle placement rather than sunlight capture, since panel tilt and direction matter more for energy output.

3. Solar Resource & Shading

The report captured a snapshot of local weather conditions at the time of assessment:

- **Temperature:** 30.1°C — within the normal operating range for solar panels, though for reference, panel efficiency typically drops slightly at higher temperatures (most panels are rated at 25°C and lose a small percentage of output for every degree above that).
 - **Cloud Cover:** 98% at the time of the scan. This is very high and, if representative of typical conditions rather than a one-off weather event, would meaningfully reduce sunlight reaching the panels and lower energy generation. The AI recommendations flag this directly, noting that "frequent cloud cover may reduce annual generation." It's worth noting this figure reflects conditions at the moment of data capture, not necessarily the annual average — but it's a signal worth monitoring, especially given Navi Mumbai's monsoon season.
 - **Wind Speed:** 17.6 km/h — a moderate breeze. Wind has a minor cooling effect on panels (which can slightly help efficiency) and is a factor structural engineers consider when specifying panel mounting hardware.
 - **Humidity:** 75% — typical for a coastal tropical location. High humidity over time can affect long-term panel and equipment durability, which is why routine maintenance (see Section 4) matters more in this climate than in drier regions.
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4. Recommended System Configuration

- **Number of Panels:** 5
 - **System Capacity:** 2.75 kW (kilowatts) — this is the combined peak power output of all panels under ideal sunlight conditions. It's a standard way to size residential systems; for context, typical Indian residential rooftop systems range from about 1 kW to 5 kW, so this falls in the small-to-mid range.
 - **Layout Strategy:** The system uses a portrait panel orientation to maximize how many panels fit into the 17.02 m² usable area, working around the two identified rooftop obstacles.
 - **Suitability:** The report explicitly notes this configuration is best suited for **residential self-consumption** — meaning the electricity generated is intended primarily to offset the property's own power usage rather than to sell back to the grid at scale.
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5. Estimated Energy Production

- **Annual Energy Output:** 4,097 kWh per year
- **Daily Energy Output:** 11.2 kWh per day (an average figure; actual daily output will vary with weather, season, and daylight hours)

To put this in context, a typical urban Indian household consumes somewhere in the range of 3,000–6,000 kWh per year depending on appliance use (especially air conditioning), so a system of this size could offset a meaningful portion — potentially all — of a modest household's annual electricity needs. Actual savings depend on local electricity tariffs and how much of the solar power is self-consumed versus exported to the grid.

6. Financial Outlook

- **Estimated Annual Savings:** ₹32,775 (based on the report's currency figures — noted here as "n" in the raw data, interpreted as Indian Rupees given the project location)
- **Payback Period:** 5.0 years — this is the time it takes for the accumulated electricity bill savings to equal the upfront cost of the system. A 5-year payback is generally considered a strong result; for reference, typical rooftop solar payback periods in India range from 4 to 8 years depending on system cost, subsidy availability, and local tariffs.
- **25-Year Profit Estimate:** ₹654,378 — this represents the total net financial benefit projected over a typical solar panel lifespan (panels are generally warranted for 25 years, though many continue producing at reduced efficiency beyond that).

These figures assume relatively stable electricity prices and consistent system performance over time; actual results will vary with future tariff changes, maintenance quality, and weather patterns.

7. Assumptions & Limitations

- This report was auto-generated and is intended as a preliminary feasibility screening, not a final engineering design.
- The weather data (temperature, cloud cover, wind, humidity) reflects conditions at the time of the automated scan and may not represent long-term seasonal averages — a full assessment should incorporate historical solar irradiance data for the site.
- The nature of the "2 rooftop obstacles" is not specified in the source data; a physical site visit is needed to confirm what they are and whether relocation is practical.
- Financial figures (savings, payback, 25-year profit) are estimates based on the system's assumed energy output and prevailing electricity rates; they do not account for potential future changes in tariffs, financing costs, or government incentive programs.
- Before proceeding, the report recommends further engineering validation, including a structural assessment of the roof and confirmation of electrical connection requirements.
- Routine maintenance is advised: cleaning panels every 2–3 months to remove dust and debris, and conducting an annual electrical and structural inspection to catch wear or damage early — this is standard practice for maintaining efficiency and safety over the system's lifetime, especially in a humid coastal climate like Navi Mumbai's.